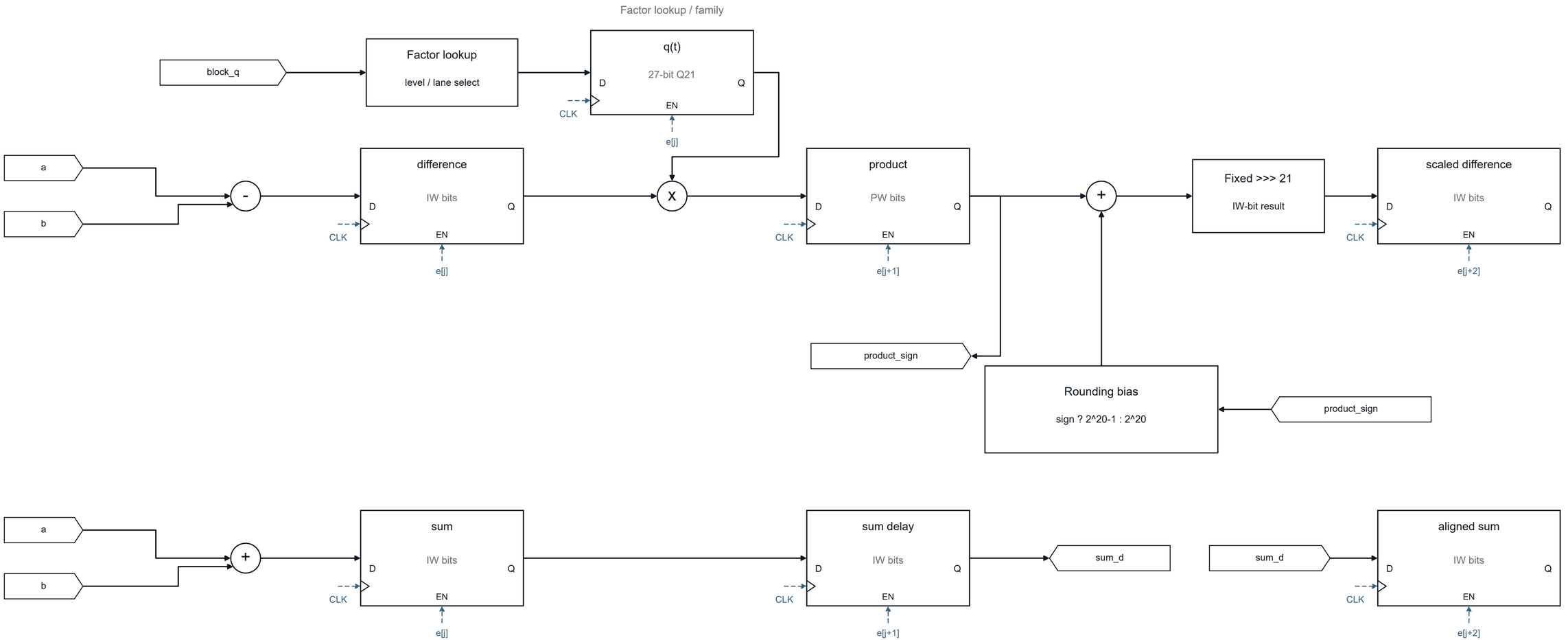
Registered butterfly / product section - no combinational MAC

Four parallel copies per product level. This section repeats at S1-S3, S4-S6 and S8-S10; C4/C8 post-adds have separate registers.



A product is registered before its rounding adder. The sum path has matching delay registers; clock-enable pauses preserve alignment.

j = 1, 4 or 8. Data registers have CLK and EN; reset is applied to the shared valid/tag pipeline shown on R01.